

Portfolio item 3: Report Skeleton

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Honours Project A

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1 Introduction

2 Supernova Ia physics and Progenitor Channels

2.1 Standardisation

- Explaining the use of type Ia supernova as standardisable candles to make distance measurements that are useful in determining cosmological parameters such as the equation of state parameter for dark energy and via determinations of H_0, Ω_m and Ω_Λ.

2.2 Progenitor Channels

- Describe the two popular progenitor channels for type Ia supernovae, the single and double degenerate scenarios and how they are differed between in the light curve data and through modelling with different nickel 56 masses and distributions.

- Describe how host galaxy properties have been used to make an inference on the delay time distributions of SNe Ia which can help determine whether these galaxy properties are associated with 'prompt' or 'delayed' SNe Ia.

2.3 Early Flux Excess Supernovae

- Here will be an explanation of how important early observation well before peak flux/magnitude of the supernova is crucial in having light curves that can help determine the progenitor channel.

- Describe the early excess flux observed in a few of the SNe Ia and how they are characterised such as through visual inspection and calculation of light curve residuals in mag and flux space.

- Discuss some of the proposed mechanisms that could explain the observed early excess in the light curves such as ^{56}Ni masses and distributions, interactions with the SNe ejecta and the companion star and interactions with the circumstellar material.

3 Host Galaxy Environment and SN Ia light curves

3.1 Host Galaxy Properties

- Discuss topics like what is known about where SNe Ia occur most radially in the Host galaxy and what bands SNe Ia trace in galaxy light to show what bands are the more likely to find them in.

- Discuss the host galaxy stellar mass step correction and proposed physical explanations for the observed correlation as an extrinsic property like host galaxy mass is not going to directly affect the brightness of the SNe Ia.

- Describe the observed differences in correlation between global properties such as specific star formation rate and the local specific star formation rate and lower masses and higher SFR in the local environment correlates with SNe Ia properties such as colour and how analysis of local environments show a homogeneity in SNe Ia properties.

3.2 The Environment and Early Excess Flux

- If early excess flux SNe Ia are found in specific environments not only will that assist in determining potential progenitor channels, it will also assist in the cutting of SNe Ia in samples used in cosmology where the early excess SNe Ia could contaminate the cosmology sample.

- The lack of a statistically significant sample size has made it so that early excess flux SNe Ia rates haven't been put as a fucntion of host galaxy properties and thus have alluded systematic classfication.

4 Fink, Rubin and LSST

- Give an overview on the specifications of LSST and Rubin telescope, including the survey strategy used for the LSST survey and describing the volume of alerts that are

expected per night. Leading to a discussion on why this volume of alerts requires a broker to process data for science.

- Describe the Fink broker and give an overview on how it helps deliver on the science it'll be used for namely the classification pipeline of alerts. Including topics like Fink's active learning approach to classifiers and the broker's performance.
- Possibly compare to the ZTF survey data that Fink has been using to the data LSST is producing and how they are different and enable new science.

5 Project Aims

- State the primary aim of the project which is to use LSST data collected via the Rubin telescope to identify and classify early excess SNe Ia and then investigate how their incidence correlates with host galaxy properties.
- Attempt to use the early excess flux SNe Ia to make constraints on possible progenitor channels.

6 Planned Methodology

- Use Fink it searches through alerts that have been classified as SN Ia candidates.
- Follow the methodology of Deckers et al. (2022) to identify the early excess flux SNe Ia along specifically around the approximate 5 days pre maximum flux and distinguishing between red and blue SNe Ia events.
- To assign SNe Ia with a host, I suspect we will use the Directional Light Radius method which takes the separation from the SN to the nearest galaxy and then normalises it by the galaxies directional light radius which is the radius of a line drawn from a galaxies centre to a nearby transient event such as a SNe Ia.
- Finally compare early excess supernovae Ia rates to host galaxies properties such as sSFR, galaxy stellar mass and colour see how colour of the SNe Ia varies across the

sample and see if any connections crop up with the host properties with colour of early excess SNe Ia. This will be used to make conclusions about whether early excess SNe Ia prefer different host galaxy environments and how that affects the possible progenitor channels for the early excess SNe Ia.

7 References

- This will be were the references are!

References

Deckers, M., Maguire, K., Magee, M. R., Dimitriadis, G., Smith, M., Murieta, A. S. d., Miller, A. A., Goobar, A., Nordin, J., Rigault, M., Bellm, E., Coughlin, M. W., Laher, R. R., Shupe, D., Graham, M. J., Kasliwal, M. M., & Walters, R. (2022). Constraining Type Ia supernova explosions and early flux excesses with the Zwicky Transient Factory [arXiv:2202.12914 [astro-ph]]. *Monthly Notices of the Royal Astronomical Society*, *512*(1), 1317–1340. https://doi.org/10.1093/mnras/stac558